FINITENESS OF ETALE COHOMOLOGY OF A CONSTANT SHEAF WITH COEFFICIENTS ANNIHILATED BY AN INTEGER PRIME TO THE CHARACTERISTIC, OR ON A PROPER CURVE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full as a chain of reductions: the prime-factor induction, the change of the coefficient ring to Lambda/ellLambda, the passage to a finite free module by descending induction on q, and the basis decomposition of underline Lambda whose cohomology is computed by commutation with direct sums.

Editorial note. The label lemma-constant-statements-coefficients names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; every reduction step and every cited input is retained exactly where the author placed it.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constant sheaf underline M on X_etale associated to a Lambda-module. Same chapter, the constant sheaf on the etale site associated to a set, abelian group or module, and its description by the etale topology. Supplies the coefficient sheaves underline M, underline Lambda, underline F_ell appearing throughout the reductions.
- Long exact cohomology sequence and induction on the exponent. Same chapter, the long exact sequence of etale cohomology attached to a short exact sequence of etale sheaves. Supplies the exact sequences used to reduce from M annihilated by n to its prime factors and from M to a free module.
- Vanishing above the dimension for the separated finite type curve. Same chapter, the vanishing and finiteness theorem for etale cohomology of tors/sheaves on separated finite type schemes of dimension <= 1 over an algebraically closed field. Supplies the vanishing in degrees > 2 used to run the descending induction on q, and the finiteness input of the same theorem.

Source reference. Original lemma 00D5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7427–7427. Prerequisite pointer invoked at proof

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line(s) 15896; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GIS: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7831–7831. Prerequisite pointer invoked at proof line(s) 15897; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03FD: sources/stacks-chow/tags.txt, source lines 4488–4488. Prerequisite pointer invoked at proof line(s) 15854; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Prerequisite pointer invoked at proof line(s) 15895; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 03SC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15656–15656. Prerequisite pointer invoked at proof line(s) 15874; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let Λ be a Noetherian ring, let M be a finite Λ-module which is annihilated by an integer $n > 0$, let k be an algebraically closed field, and let X be a separated, finite type scheme of dimension ≤ 1 over k. Then*

(1) *$H^q_{étale}(X, \underline{M})$ is a finite Λ-module if n is prime to char(k),*
(2) *$H^q_{étale}(X, \underline{M})$ is a finite Λ-module if X is proper.*

Proof. If $n = \ell n'$ for some prime number ℓ, then we get a short exact sequence $0 \to M[\ell] \to M \to M' \to 0$ of finite Λ-modules and M' is annihilated by n'. This produces a corresponding short exact sequence of constant sheaves, which in turn gives rise to an exact sequence of cohomology modules

$$H^q_{étale}(X, \underline{M[n]}) \to H^q_{étale}(X, \underline{M}) \to H^q_{étale}(X, \underline{M'})$$

Thus, if we can show the result in case M is annihilated by a prime number, then by induction on n we win.

Let ℓ be a prime number such that ℓ annihilates M. Then we can replace Λ by the $\mathbf{F}_\ell$-algebra $\Lambda/\ell\Lambda$. Namely, the cohomology of $\mathcal{F}$ as a sheaf of Λ-modules is the same as the cohomology of $\mathcal{F}$ as a sheaf of $\Lambda/\ell\Lambda$-modules, for example by Cohomology on Sites, Lemma 03FD.

Assume ℓ be a prime number such that ℓ annihilates M and Λ. Let us reduce to the case where M is a finite free Λ-module. Namely, choose a short exact sequence

$$0 \to N \to \Lambda^{\oplus m} \to M \to 0$$

This determines an exact sequence

$$H^q_{étale}(X, \underline{\Lambda^{\oplus m}}) \to H^q_{étale}(X, \underline{M}) \to H^{q+1}_{étale}(X, \underline{N})$$

By descending induction on q we get the result for M if we know the result for $\Lambda^{\oplus m}$. Here we use that we know that our cohomology groups vanish in degrees > 2 by Theorem 03SC.

Let ℓ be a prime number and assume that ℓ annihilates Λ. It remains to show that the cohomology groups $H^q_{étale}(X, \underline{\Lambda})$ are finite Λ-modules. We will use a trick to show this; the "correct" argument uses a coefficient theorem which we will show later. Choose a basis $\Lambda = \bigoplus_{i \in I} \mathbf{F}_\ell e_i$ such that $e_0 = 1$ for some $0 \in I$. The choice of this basis determines an isomorphism

$$\underline{\Lambda} = \bigoplus \underline{\mathbf{F}_\ell} e_i$$

of sheaves on $X_{étale}$. Thus we see that

$$H^q_{étale}(X, \underline{\Lambda}) = H^q_{étale}(X, \bigoplus \underline{\mathbf{F}_\ell} e_i) = \bigoplus H^q_{étale}(X, \underline{\mathbf{F}_\ell}) e_i$$

since taking cohomology over X commutes with direct sums by Theorem 09YQ (or Lemma 00D5 or Lemma 0GIS). Since we already know that $H^q_{étale}(X, \underline{\mathbf{F}_\ell})$ is a finite dimensional $\mathbf{F}_\ell$-vector space (by Theorem 03SC), we see that $H^q_{étale}(X, \underline{\Lambda})$ is free over Λ of the same rank. Namely, given a basis $\xi_1, \ldots, \xi_m$ of $H^q_{étale}(X, \underline{\mathbf{F}_\ell})$ we see that $\xi_1 e_0, \ldots, \xi_m e_0$ form a Λ-basis for $H^q_{étale}(X, \underline{\Lambda})$. □